The Bayesian Estimation of a Linear Functional Relationship

By D. V. LINDLEY and G. M. EL-SAYYAD

University College of Wales, Aberystwyth†

[Received March 1967. Revised July 1967]

SUMMARY

The purpose of this note is twofold. First, to demonstrate by means of an example that a statistical approach based solely on the likelihood function can be misleading, whereas the Bayesian method yields sensible results. Second, to develop the Bayesian method and apply the results to the problem of estimating a linear functional relationship. In Section 1 we discuss the likelihood approach. Section 2 concerns the Bayesian method, which is applied to the functional relationship situation in Section 3: it is shown that the asymptotic behaviour of the distributions exhibits a novel form. In Section 4 the posterior distributions derived in Section 3 are considered in more detail: in particular, a standard error for the estimate of slope is provided. Section 5 contains various comments on the method, and, in particular, indicates other fields of application.

1. THE LIKELIHOOD APPROACH

THE situation we consider has been discussed by many authors: convenient references are Kendall and Stuart (1961, Chapter 29), Johnston (1963, Chapter 6) and Madansky (1959). We use the language of the first reference and speak of a *functional* relationship between mathematical quantities, but do not use their notation because we prefer, in a Bayesian approach, consistently to use Greek letters for parameters. There are two quantities which are linearly related and it is desired to estimate the relationship. Unfortunately, observations on both the quantities are subject to error and the estimation has to be based on such observations. Formally, ξ and η satisfy

$$\eta = \theta\xi, \tag{1}$$

where θ is unknown. For any value of ξ, ξ_i say, x_i is an observation normally distributed about ξ_i with variance ϕ_1; y_i is an observation normally distributed about $\eta_i = \theta\xi_i$ with variance ϕ_2. For given ξ_i these observations are independent, and independent of other observations at different values of ξ. It is desired to estimate θ on the basis of observations $(x_1, y_1; x_2, y_2; \ldots; x_n, y_n)$: $\phi_1, \phi_2, \xi_1, \ldots, \xi_n$ (and of course θ) are all unknown. Notice that in (1) it has been supposed that the relationship passes through the origin. We could treat the more general linear situation, $\eta = \alpha + \theta\xi$, but to do so would complicate the argument and notation. The points we wish to emphasize are not affected by the restriction $\alpha = 0$. The general case can be treated by regarding the x's and y's in what follows as deviations from their sample means. There is no difficulty in estimating α and the loss of one degree of freedom will scarcely matter, especially since we shall be concerned with large or, at any rate, not very small samples. We do not assume the sign of θ known. In many applications it will be known that θ is positive. We comment on this later (before equation (23)).

† Present addresses: D.V.L.: University College, London. G.M.E.: University of Riyadh, Saudi Arabia.

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

Now it is well known that the likelihood approach to this problem gets into difficulties. We repeat the salient features here in order to compare them with the Bayesian results: for details we refer the reader to the references already cited. The likelihood function is proportional to

$$(\phi_1\phi_2)^{-\frac{1}{2}n}\exp\{-\tfrac{1}{2}\Sigma(x_i-\xi_i)^2/\phi_1-\tfrac{1}{2}\Sigma(y_i-\theta\xi_i)^2/\phi_2\}. \tag{2}$$

(All summations and products in this paper are from 1 to n.) The unknown parameters are $(n+3)$ in number; namely $\theta, \phi_1, \phi_2, \xi_1, \ldots, \xi_n$. In the language of Neyman and Scott (1951) the parameters θ, ϕ_1 and ϕ_2 are "structural", whereas the ξ's are "incidental"; the distinction being that the structural parameters occur in the joint distribution of every observation pair, whereas ξ_i is incidental to the pair (x_i, y_i) alone. Differentiating with respect to each of the parameters in turn and equating the results to zero, we obtain the following maximum-likelihood equations:

$$\Sigma(y_i-\hat{\theta}\hat{\xi}_i)\,\hat{\xi}_i = 0, \tag{3a}$$

$$n^{-1}\Sigma(x_i-\hat{\xi}_i)^2 = \hat{\phi}_1, \tag{3b}$$

$$n^{-1}\Sigma(y_i-\hat{\theta}\hat{\xi}_i)^2 = \hat{\phi}_2, \tag{3c}$$

and

$$(x_i-\hat{\xi}_i)/\hat{\phi}_1+(y_i-\hat{\theta}\hat{\xi}_i)\,\hat{\theta}/\hat{\phi}_2 = 0. \tag{3d}$$

If (3d) is solved for $\hat{\xi}_i$ we obtain the results

$$(x_i-\hat{\xi}_i) = (x_i\,\hat{\theta}-y_i)\,\hat{\phi}_1\,\hat{\theta}/(\hat{\phi}_2+\hat{\phi}_1\,\hat{\theta}^2)$$

and

$$-(y_i-\hat{\theta}\hat{\xi}_i) = (x_i\,\hat{\theta}-y_i)\,\hat{\phi}_2/(\hat{\phi}_2+\hat{\phi}_1\,\hat{\theta}^2).$$

Substitution of these two results into (3b) and (3c) gives

$$\hat{\theta} = (\hat{\phi}_2/\hat{\phi}_1)^{\frac{1}{2}}. \tag{4}$$

This is a disturbing result, for it says that the maximum-likelihood estimate of the slope is equal to the estimate of the ratio of the standard deviations of the two errors of observation in y and x. If, for example, the estimation of θ were consistent, the estimation of either ϕ_1 or ϕ_2 would be inconsistent save in the exceptional case when (4) held for the true values. It is relevant to later considerations to remark that $\hat{\theta}$ does converge in probability to a value as $n\to\infty$: in fact, simple calculation shows that $\hat{\theta} = (\Sigma y_i^2/\Sigma x_i^2)^{\frac{1}{2}}$ and, if $\Sigma\xi_i^2/n$ converges to V, say, as $n\to\infty$, $\hat{\theta}$ converges to $\{(\theta^2 V+\phi_2)/(V+\phi_1)\}^{\frac{1}{2}}$, which only equals θ if $\theta = \{\phi_2/\phi_1\}^{\frac{1}{2}}$ (compare equation (4)). Consequently, apart from the exceptional case, $\hat{\theta}$ converges to the wrong value. Ordinary maximum-likelihood estimation fails. Any more sophisticated treatment of the likelihood function is fraught with difficulties since it is a function in $(n+3)$ dimensions. This is a puzzling situation since anyone who has had experience of data that appear to fit the proposed model reasonably well knows that a naive look at the data suggests that something can be said about the functional relationship. He is often sceptical of the statistician who says that estimation is not possible. In our view this scepticism is well founded, and it is the maximum-likelihood method that is at fault. We now reconsider the problem using Bayesian methods.

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

2. THE BAYESIAN APPROACH

Since, in a Bayesian approach, all variables are random; that is, have a probability distribution, it is important to formulate one's probability model with some care. It is perhaps most convenient to do this in the general case of incidental and structural parameters before specializing to the functional relationship case. This will also have the advantage of demonstrating the generality of the method.

The incidental parameters will be denoted by $\boldsymbol{\xi} = (\xi_1, \xi_2, \ldots, \xi_n)$ and the structural parameters by $\boldsymbol{\theta}$. The parameters will have a (prior) probability distribution. In the case of the incidental parameters this distribution can most easily be expressed by supposing it to depend on other parameters, which we shall see can be estimated. Denote these parameters by $\boldsymbol{\tau}$. The observations will be written $\mathbf{z} = (z_1, z_2, \ldots, z_n)$. Any of these quantities, including z_i and ξ_i, may be vector-valued. We assume

(1) $\boldsymbol{\theta}$ and $\boldsymbol{\tau}$ have a joint (prior) distribution $p(\boldsymbol{\theta}, \boldsymbol{\tau})$,

(2) the ξ_i, given $\boldsymbol{\theta}$ and $\boldsymbol{\tau}$, are independent: that is,

$$p(\boldsymbol{\xi}|\boldsymbol{\theta}, \boldsymbol{\tau}) = \prod p(\xi_i|\boldsymbol{\theta}, \boldsymbol{\tau}), \tag{5}$$

(3) the z_i, given $\boldsymbol{\xi}$, $\boldsymbol{\theta}$ and $\boldsymbol{\tau}$ are independent: that is,

$$p(\mathbf{z}|\boldsymbol{\xi}, \boldsymbol{\theta}, \boldsymbol{\tau}) = \prod p(z_i|\boldsymbol{\xi}, \boldsymbol{\theta}, \boldsymbol{\tau}), \tag{6}$$

(4) for given ξ_i, $\boldsymbol{\theta}$ and $\boldsymbol{\tau}$; z_i is independent of all ξ_j with $j \neq i$: that is

$$p(z_i|\boldsymbol{\xi}, \boldsymbol{\theta}, \boldsymbol{\tau}) = p(z_i|\xi_i, \boldsymbol{\theta}, \boldsymbol{\tau}).$$

Assumption (1) is simply the statement of the prior distribution, and (2) merely expresses the belief that knowledge of one incidental parameter would not affect knowledge of the rest, given that one knew the value of $\boldsymbol{\tau}$. In effect, $\boldsymbol{\tau}$ provides correlation between the ξ_i, which would naturally exist in practice. Assumption (3) is the usual assumption about the independence of the observations, and (4) expresses the incidental nature of each ξ_i, affecting only z_i. Assumptions (3) and (4) are part of the likelihood approach. Notice that the four assumptions enable the complete probability distribution of $\mathbf{z}$, $\boldsymbol{\xi}$, $\boldsymbol{\theta}$ and $\boldsymbol{\tau}$ to be specified. This is accomplished by a sequence of conditional distributions: first $\boldsymbol{\theta}$ and $\boldsymbol{\tau}$; then $\boldsymbol{\xi}$ for fixed $\boldsymbol{\theta}$ and $\boldsymbol{\tau}$; then $\mathbf{z}$ for fixed $\boldsymbol{\theta}$, $\boldsymbol{\tau}$ and $\boldsymbol{\xi}$. This is usually the most systematic and convenient way of describing the distributions, rather than by trying to express a joint distribution for all the parameters.

We can now eliminate the incidental parameters.

For

$$\begin{aligned}
p(\mathbf{z}|\boldsymbol{\theta}, \boldsymbol{\tau}) &= \int p(\mathbf{z}, \boldsymbol{\xi}|\boldsymbol{\theta}, \boldsymbol{\tau})\, d\boldsymbol{\xi} \\
&= \int p(\mathbf{z}|\boldsymbol{\xi}, \boldsymbol{\theta}, \boldsymbol{\tau})\, p(\boldsymbol{\xi}|\boldsymbol{\theta}, \boldsymbol{\tau})\, d\boldsymbol{\xi} \\
&= \int \prod p(z_i|\boldsymbol{\xi}, \boldsymbol{\theta}, \boldsymbol{\tau})\, p(\xi_i|\boldsymbol{\theta}, \boldsymbol{\tau})\, d\boldsymbol{\xi} \quad \text{(by assumptions (2) and (3))} \\
&= \prod \int p(z_i|\xi_i, \boldsymbol{\theta}, \boldsymbol{\tau})\, p(\xi_i|\boldsymbol{\theta}, \boldsymbol{\tau})\, d\xi_i \quad \text{(by assumption (4))} \\
&= \prod p(z_i|\boldsymbol{\theta}, \boldsymbol{\tau}).
\end{aligned} \tag{7}$$

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

We therefore see that, given $\boldsymbol{\theta}$ and $\boldsymbol{\tau}$, the z_i are independent random variables with a common distribution. The posterior distribution of $\boldsymbol{\theta}$ and $\boldsymbol{\tau}$, and hence of $\boldsymbol{\theta}$, may be found by the usual use of Bayes formula; and, moreover, will have the usual properties associated with posterior distributions derived from a sequence of independent, identically distributed observations, provided the underlying distribution $p(z_i|\boldsymbol{\theta},\boldsymbol{\tau})$ is not degenerate. For example, the posterior distribution will concentrate, as $n\to\infty$, around the true value. We have therefore removed the incidental parameters completely and reduced the problem to a standard one involving a fixed set, the structural parameters, under the usual conditions of a random sample.

Kiefer and Wolfowitz (1956) similarly introduced a distribution for the incidental parameters and proved that (7) provides estimates of $\boldsymbol{\theta}$ and $\boldsymbol{\tau}$. But they regarded (7) as the likelihood function and presumably thought of (5) as a frequency probability. In our approach (5) is an expression of beliefs about the incidental parameters and does not imply that the ξ_i have this as a frequency distribution. Of course, our final posterior beliefs depend on this assumption, as we shall see most strikingly in the functional relationship situation below. Nevertheless the mathematical results of Kiefer and Wolfowitz are highly relevant for us. They discuss the maximum-likelihood estimates based on (7) rather than any Bayes estimate. The former may be unsatisfactory even in the independent, identically distributed case, whereas the latter are always satisfactory for a proper prior. Their paper (in its final paragraph) contains an elegant, non-pathological example of the failure of maximum likelihood that deserves to be better known. We shall go further than these authors and provide explicit approximations for the estimates and their standard errors.

3. THE FUNCTIONAL RELATIONSHIP

The method of the previous section will now be applied to the functional relationship situation of Section 1. The structural parameter $\boldsymbol{\theta}$ is the triplet (θ,ϕ_1,ϕ_2). We shall particularize assumption (2) above to read that given θ, ϕ_1, ϕ_2 and τ, the ξ_i are independent $N(0,\tau)$. The (prior) distribution of $\boldsymbol{\theta}$ and τ will be written $\pi(\theta,\phi_1,\phi_2,\tau)$. Assumptions (3) and (4) have already been discussed in connection with the likelihood in Section 2, the observations z_i being the pair (x_i,y_i). The distribution

$$p(x_i,y_i|\,\theta,\phi_1,\phi_2,\tau)$$

of an observation pair given the structural parameters (and τ) can easily be written down by remarking that $x_i=\xi_i+\epsilon_{i1}$, $y_i=\theta\xi_i+\epsilon_{i2}$, where ϵ_{ij} are independent $N(0,\phi_j)$ $(j=1,2)$ and, by the assumption just made, ξ_i are $N(0,\tau)$. Consequently, (x_i,y_i) have a bivariate normal distribution with zero means and

$$\left.\begin{aligned}\operatorname{var}(x_i|\,\theta,\phi_1,\phi_2,\tau)&=\tau+\phi_1,\\ \operatorname{cov}(x_i,y_i|\,\theta,\phi_1,\phi_2,\tau)&=\theta\tau\\ \text{and}\qquad \operatorname{var}(y_i|\,\theta,\phi_1,\phi_2,\tau)&=\theta^2\tau+\phi_2.\end{aligned}\right\}\tag{8}$$

Denote these moments by θ_{11}, θ_{12} and θ_{22} respectively. A random sample from a bivariate normal distribution with zero means has a set of minimal sufficient statistics, namely $n^{-1}\sum x_i^2=s_{11}$, $n^{-1}\sum x_iy_i=s_{12}$, $n^{-1}\sum y_i^2=s_{22}$, say. Consequently, on combining (7) with the prior distribution of the structural parameters (and τ) we obtain, by Bayes formula, the posterior distribution to be proportional to

$$\frac{\pi(\theta,\phi_1,\phi_2,\tau)}{(\phi_1\phi_2+\tau\phi_2+\tau\phi_1\theta^2)^{\frac{1}{2}n}}\exp\left\{-\tfrac{1}{2}n\frac{s_{11}(\theta^2\tau+\phi_2)-2s_{12}\theta\tau+s_{22}(\tau+\phi_1)}{\phi_1\phi_2+\tau\phi_2+\tau\phi_1\theta^2}\right\}.\tag{9}$$

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

Further simplification is possible if we transform from the four parameters $\theta, \phi_1, \phi_2, \tau$ to new parameters consisting of the variances and covariances given by (8) plus θ, which is left unaltered since it is the parameter of principal interest to us. The Jacobian of the transformation is $|\theta|$ and we obtain for the joint posterior density of θ, θ_{11}, θ_{12} and θ_{22}

$$\frac{K\pi(\theta, \theta_{11}-\theta_{12}/\theta, \theta_{22}-\theta\theta_{12}, \theta_{12}/\theta)}{|\theta|(\theta_{11}\theta_{22}-\theta_{12}^2)^{\frac{1}{2}n}} \exp\left(-\tfrac{1}{2}n\frac{s_{11}\theta_{22}-2s_{12}\theta_{12}+s_{22}\theta_{11}}{\theta_{11}\theta_{22}-\theta_{12}^2}\right), \qquad (10)$$

where K is a constant.

Parts of this distribution depend on n, the sample size: namely, the exponential term and the power of $\theta_{11}\theta_{22}-\theta_{12}^2$ in the denominator. The remainder does not involve n. The remarkable feature of the distribution is that the parts which do involve n do not involve θ, the parameter of interest. The consequences of this are important. If (10) is integrated with respect to θ to obtain the marginal distribution of the variances and covariances, θ_{11}, θ_{12} and θ_{22}, we shall obtain a result similar to (10), namely a part, equal to that in (10), which involves n, and another which does not. This distribution tends to concentrate as n tends to infinity and the values of θ_{ij} become known with certainty. (That this is so can be seen by treating θ_{ij} as the elements of a 2×2 matrix and transforming to θ^{ij} the elements of the inverse matrix. The θ^{ij} will be seen to have a limiting Wishart distribution (Anderson, 1958) and the known properties of that distribution establish the result.) The values around which the distribution of θ_{ij} concentrates are s_{ij}. In other words, the sample moments estimate the population moments, equation (8). The behaviour of the distribution of θ_{ij} is therefore typical of posterior distributions with large samples.

The distribution of θ will behave quite differently. Let us consider first very large samples and crude approximations (better approximations are considered in the next section). In (10) θ_{ij} can then effectively be replaced by s_{ij} and the posterior distribution of θ is approximately

$$K\pi(\theta, s_{11}-s_{12}/\theta, s_{22}-\theta s_{12}, s_{12}/\theta)|\theta|^{-1}. \qquad (11)$$

This distribution does not concentrate around any value as $n\to\infty$ and the variance, for example, does not tend to zero. Consequently, however large the sample, we never learn the true value of θ but only obtain a distribution for it of non-zero dispersion. This is in marked contrast to the maximum-likelihood approach outlined in Section 1 where the dispersion of the estimate did tend to zero, but the result was inconsistent. This is not to say we do not learn a lot about θ; typically, we do. But such knowledge as we gain depends heavily on the prior distribution of all the parameters (and not just θ). Indeed all our knowledge of θ gained from the data comes through our prior beliefs of how θ is related to the variances and covariances θ_{ij}, equation (8). It will typically be true that these beliefs are quite strong, in view of the relations between θ_{ij} and θ. For example, one immediate remark that can be made is that ϕ_1 and ϕ_2 are *a priori* both non-negative so that, from (11), with probability one,

$$s_{11}-s_{12}/\theta \geqslant 0 \quad \text{and} \quad s_{22}-s_{12}\theta \geqslant 0,$$

giving

$$s_{12}/s_{11} \leqslant \theta \leqslant s_{22}/s_{12} \qquad (12)$$

if $s_{12}>0$, and the opposite inequalities in the contrary case. Inequalities equivalent to these have been noted by Moran (1956). The distribution of θ is therefore confined to this interval whatever be the prior.

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

To appreciate the distribution (11) a little more let us consider a typical form for the prior density π. In many applications it would not be unreasonable to assume θ, τ and the pair (ϕ_1, ϕ_2) to be mutually independent. This independence would imply quite strong dependence between the θ_{ij} and θ and hence appreciable knowledge about θ, at least in large samples. For example, suppose $\tan^{-1}\theta$ is uniformly distributed in the interval $[-\frac{1}{2}\pi, \frac{1}{2}\pi)$ and the usual vague prior for τ with $\log\tau$ uniform is assumed: then the posterior density is

$$K\pi^*(s_{11}-s_{12}/\theta, s_{22}-s_{12}\theta)/(1+\theta^2), \tag{13}$$

where π^* is the joint prior for ϕ_1 and ϕ_2. π^* will be zero outside the range (12). If ϕ_1 and ϕ_2 are independent and unimodal the distribution of θ will tend to be bimodal with local maxima at values of θ given by $s_{11}-s_{12}/\theta = \phi_1^*$ and $s_{22}-s_{12}\theta = \phi_2^*$, where ϕ_i^* are the modes for ϕ_i. Another possibility, which we explore in detail in the next section, is to assume ϕ_1 and $\lambda = \phi_2/\phi_1$ to be independent.

The densities, (11) and (13), will only be useful in extremely large samples, when θ_{ij} is effectively known to be s_{ij}. We now consider better approximations and defer until a later section further general comments on the method and its results.

4. The Posterior Distributions

Our aim in this section is to explore the joint posterior distribution of θ, ϕ_1, ϕ_2 and τ (equation (9)) more thoroughly than we did in the preceding section in order to produce manageable approximations for the posterior distribution of θ. It will still be necessary to assume large samples, but we feel the approximations are better than the crude ones used above, and that therefore the samples need probably only be of moderate size to make the results sensible. Our approach is to work in terms of θ, ϕ, λ and τ, where $\phi = \phi_1$ and $\lambda = \phi_2/\phi_1$ is the ratio of the error variances, and proceed first to find the posterior distribution of θ, given λ. This we feel is a result of interest in its own right in view of the numerous studies which have been made of the case λ known in default of solutions, by the likelihood approach, to the general case. A final integration with respect to λ completes the result.

In (9) make the substitutions $\lambda = \phi_2/\phi_1$ and $\rho = \tau/\phi_1$, and write ϕ instead of ϕ_1. The posterior density of θ, ϕ, λ and ρ is then

$$\frac{\pi(\theta,\phi,\lambda,\rho)}{\phi^n(\lambda+\lambda\rho+\rho\theta^2)^{\frac{1}{2}n}}\exp\left\{-\tfrac{1}{2}n\frac{s_{11}(\theta^2\rho+\lambda)-2s_{12}\theta\rho+s_{22}(\rho+1)}{\phi(\lambda+\lambda\rho+\rho\theta^2)}\right\} \tag{14}$$

apart from a constant of proportionality, dependent only on the data. Here $\pi(\theta, \phi, \lambda, \rho)$ is the prior distribution of these four parameters: explicit forms for parts of this will be introduced as is necessary in the course of the arguments. As indicated in the previous section, some facets of it will be important, even in large samples: others can be treated with less precision, as is usual in most statistical problems. If we assume the usual "ignorance" distribution for ϕ, proportional to ϕ^{-1}, independent of the other parameters, the integration with respect to ϕ in (14) can easily be performed exactly, with the result that the posterior density of θ, λ and ρ is proportional to

$$\frac{\pi(\theta,\lambda,\rho)}{(\lambda+\lambda\rho+\rho\theta^2)^{\frac{1}{2}n}}\left\{\frac{\lambda+\lambda\rho+\rho\theta^2}{s_{11}(\theta^2\rho+\lambda)-2s_{12}\theta\rho+s_{22}(\rho+1)}\right\}^n$$

$$= \pi(\theta,\lambda,\rho)\left[\frac{\rho(\lambda+\theta^2)+\lambda}{\{\rho(s_{11}\theta^2-2s_{12}\theta+s_{22})+\lambda s_{11}+s_{22}\}^2}\right]^{\frac{1}{2}n}. \tag{15}$$

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

We next consider the integration with respect to ρ. This is of the form

$$\int \pi(\theta, \lambda, \rho)\{(a\rho+b)/(c\rho+d)^2\}^{\frac{1}{2}n}\, d\rho,$$

where a, b, c and d do not depend on ρ. If we write $f(\rho) = \log(c\rho+d)^2/(a\rho+b)$, the usual steepest descent argument gives as an approximation to this integral

$$\pi(\theta, \lambda, \rho_0)\{4\pi/nf''(\rho_0)\}^{\frac{1}{2}} \exp\{-\tfrac{1}{2}nf(\rho_0)\},$$

where ρ_0 is the value of ρ for which $f(\rho)$ is a minimum. Simple calculation shows that here

$$\left.\begin{aligned} \rho_0 &= d/c - 2b/a, \\ f(\rho_0) &= \log 4\Delta c/a^2 \\ \text{and} \qquad f''(\rho_0) &= a^2c^2/2\Delta^2, \end{aligned}\right\} \tag{16}$$

where $\Delta = ad-bc = s_{22}\theta^2 + 2s_{12}\lambda\theta + \lambda^2 s_{11}$. This argument assumes $\rho_0 > 0$ since necessarily $\rho > 0$. Consequently, when $\rho_0 > 0$ the posterior density of θ and λ is approximately proportional to

$$\pi(\theta, \lambda, \rho_0) \cdot \frac{(\lambda+\theta^2)^{n-1}}{(s_{22}\theta^2 + 2s_{12}\lambda\theta + \lambda^2 s_{11})^{\frac{1}{2}n-1}(s_{11}\theta^2 - 2s_{12}\theta + s_{22})^{\frac{1}{2}n+1}}. \tag{17}$$

The above argument is equivalent to replacing the distribution of ρ, for fixed θ and λ, by a normal distribution with the same mode and second logarithmic derivative at the mode. We use this method again, without the subsequent integration, to obtain from (17) an approximation to the distribution of θ for fixed λ. At first it appears difficult to obtain the value of θ which maximizes (17). The difficulty can be overcome by writing (17) in the form

$$\pi^*(\theta, \lambda)\{(Q_0/Q_1)(Q_0/Q_2)\}^{\frac{1}{2}n}, \tag{18}$$

where Q_i $(i = 0, 1, 2)$ are three quadratics in θ, namely

$$\left.\begin{aligned} Q_0 &= \theta^2 + \lambda, \\ Q_1 &= s_{22}\theta^2 + 2s_{12}\lambda\theta + \lambda^2 s_{11} \\ \text{and} \qquad Q_2 &= s_{11}\theta^2 - 2s_{12}\theta + s_{22}, \end{aligned}\right\}$$

and

$$\pi^*(\theta, \lambda) = \pi(\theta, \lambda, \rho_0)\, Q_1/Q_0\, Q_2.$$

It follows that the maximum is approximately attained at the maximum of $(Q_0/Q_1)(Q_0/Q_2)$. It is easily verified that $Q_1 + \lambda Q_2 = (s_{22} + \lambda s_{11})\, Q_0$ and hence we have to minimize $(s_{22} + \lambda s_{11} - \lambda Q_2/Q_0)(Q_2/Q_0)$. This is a quadratic function of Q_2/Q_0 and hence equivalently we require the maximum of $|Q_2/Q_0 - (s_{22} + \lambda s_{11})/2\lambda|$. In addition it must be remembered that $\rho_0 > 0$, and, from (16), the condition for this is easily seen to be $Q_2/Q_0 - (s_{22} + \lambda s_{11})/2\lambda < 0$. So finally we require the least (negative) value of $Q_2/Q_0 - (s_{22} + \lambda s_{11})/2\lambda$. Simple differentiation shows that this occurs when θ is the root of the equation

$$\theta^2 + t\theta - \lambda = 0 \tag{19}$$

with the same sign as s_{12}, where

$$t = (\lambda s_{11} - s_{22})/s_{12}. \tag{20}$$

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

For large n, the effect of the remaining term $\pi^*(\theta, \lambda)$ on the maximum of (18) will be negligible and hence the posterior distribution of θ, for fixed λ, has its maximum at the root of (19), at least approximately.

It remains to find the second logarithmic derivative at the maximum. This is $\frac{1}{2}n$ times

$$2\frac{Q_0 Q_0'' - Q_0'^2}{Q_0^2} - \frac{Q_1 Q_1'' - Q_1'^2}{Q_1^2} - \frac{Q_2 Q_2'' - Q_2'^2}{Q_2^2}.$$

But, in the usual way, at the maximum $(Q_0'/Q_0)^2 = (Q_1'/Q_1)^2 = (Q_2'/Q_2)^2$, so that this becomes simply

$$2Q_0''/Q_0 - Q_1''/Q_1 - Q_2''/Q_2 = \frac{4}{\theta^2+\lambda} - \frac{2s_{22}}{s_{22}\theta^2 + 2s_{12}\lambda\theta + \lambda^2 s_{11}} - \frac{2s_{11}}{s_{11}\theta^2 - 2s_{12}\theta + s_{22}}.$$

Again using the same result in the form $Q_i = Q_i' Q_0/Q_0'$ $(i = 1, 2)$ at the maximum, to eliminate Q_1, Q_2, this is

$$\begin{aligned}
&\frac{2}{\lambda+\theta^2}\left(2 - \frac{\theta s_{22}}{\theta s_{22} + s_{12}\lambda} - \frac{\theta s_{11}}{\theta s_{11} - s_{12}}\right) \\
&\quad = \frac{2}{\lambda+\theta^2}\,\frac{s_{12}\{\theta(\lambda s_{11} - s_{22}) - 2s_{12}\lambda\}}{\theta^2 s_{11}s_{22} + \theta s_{12}(\lambda s_{11} - s_{22}) - s_{12}^2\lambda} \\
&\quad = \frac{-2}{\lambda+\theta^2}\,\frac{s_{12}(\theta^2 s_{12} + \lambda s_{12})}{\theta^2(s_{11}s_{22} - s_{12}^2)}, \quad \text{from (19)} \\
&\quad = -\frac{2}{\theta^2}\,\frac{s_{12}^2}{s_{11}s_{22} - s_{12}^2}.
\end{aligned}$$

Consequently, we have the following result: *in large samples the approximate posterior distribution of θ for fixed λ is normal with mean θ_λ, the root of* (19) *with the sign of s_{12}, and variance*

$$\theta_\lambda^2 u^2/n, \tag{21}$$

where

$$u^2 = s_{11}s_{22}/s_{12}^2 - 1 = (1 - r^2)/r^2,$$

r being the sample correlation coefficient.

In (21) θ_λ is the usual estimate of θ provided by least squares and maximum likelihood when λ is known. This is an example of the usual situation in large samples, where the maximum likelihood and posterior mode estimates coincide. It is also typically true that the sampling variance of the maximum-likelihood estimate agrees, again in large samples, with the variance of the posterior distribution. The former has been found approximately by Madansky (1959) using results obtained by Creasy (1956). We have compared his expression with the posterior variance (21) and found that they do agree provided some reasonable approximations are made. Madansky's expression is somewhat complicated, but using such devices as replacing θ by θ_λ (which one would have to do in any application of his result) it reduces to the simpler form (21). We do not weary the reader with details. It follows that for fixed λ the

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

Bayesian result agrees with the maximum-likelihood one. The reader who still adheres to the likelihood method may find our new expression for the sampling variance to be of assistance.

Madansky used the model described in Section 1 with the ξ_i fixed. The referee points out that if we use the assumption we have made that the ξ_i are $N(0, \tau)$, then the large sample variance of the statistic θ_λ, given θ, ϕ_1, ϕ_2 and τ, may be calculated and found to agree with (21) when estimates are substituted for the parameters. The calculation may be performed by first finding the large sample variances and covariances of s^{11}, s^{12}, s^{22}, the maximum-likelihood estimates of the elements of the inverse of the variance–covariance matrix given by (8); by inverting the information matrix and then using this to obtain the large sample variance of t, defined by (20), which is a simple rational function of s^{11}, s^{12}, s^{22}. Finally, from (19), in an obvious notation

$$E(2\theta_\lambda + t)\,\delta\theta_\lambda + E(\theta_\lambda)\,\delta t = 0$$

and $\text{var}(\theta_\lambda)$ may be related to $\text{var}(t)$.

It is worthy of note that the coefficient of variation of θ is $\{(1-r^2)/nr^2\}^{\frac{1}{2}}$, which is the same as the coefficient of variation of the estimate of the regression coefficient of y on x, with the sample correlation replacing the population correlation; as is necessary in any use of the standard error of an estimate.

The argument just given refers to the distribution of θ for fixed λ: we now return to the joint density (17). We first evaluate (17) when $\theta = \theta_\lambda$. Yet again using the result $Q_i = Q_i' Q_0 / Q_0'$ $(i = 1, 2)$ at the maximum to eliminate Q_1 and Q_2, we have for (17) with $\theta = \theta_\lambda$

$$\frac{\pi(\theta, \lambda, \rho_0)}{(\lambda + \theta^2)} \left(\frac{\theta s_{22} + s_{12}\lambda}{\theta s_{11} - s_{12}}\right)^{\frac{1}{2}} \frac{\theta^n}{\{\theta^2 s_{11} s_{22} + \theta s_{12}(\lambda s_{11} - s_{22}) - s_{12}^2 \lambda\}^{\frac{1}{2}n}}$$

$$= \frac{\pi(\theta, \lambda, \rho_0)}{(\lambda + \theta^2)} \left(\frac{\theta s_{22} + s_{12}\lambda}{\theta s_{11} - s_{12}}\right)^{\frac{1}{2}} \frac{1}{(s_{11} s_{22} - s_{12}^2)^{\frac{1}{2}n}}. \qquad (22)$$

(The reader may compare this with the argument leading to (21).) The important feature of this result is that the term involving n does not depend on θ or λ. Consequently, as n tends to infinity the joint distribution does not concentrate around any value and, in particular, we may not use the approximations based on the maximum and the second logarithmic derivative there that have been used in the earlier part of the argument. This effective degeneracy in the joint distribution of θ and λ is, of course, just what we should expect from the considerations in Section 3. It follows that the posterior distribution of θ does not have a variance which tends to zero as n increases. This distribution can be obtained by taking the distribution of θ for fixed λ and integrating with respect to the distribution of λ. We therefore see that prior opinions about λ are critical and we cannot, as we have done so far, argue that the prior distribution plays a small role in large samples and that therefore it suffices to assume it smooth in the neighbourhood of the maximum.

We therefore consider a special form for the prior distribution. It has already been supposed that ϕ has prior density proportional to ϕ^{-1}, independent of the other parameters (but see Section 5 below). We similarly suppose that θ and ρ have independent prior densities which are constant and proportional to ρ^{-1} respectively, independent of λ whose prior distribution we shall leave as $\pi(\lambda)$. If the sign of θ were known another prior might be appropriate. This will modify $\pi^*(\lambda)$ below.

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

With uniform prior it will then follow from (17), using the conditional distribution of θ given λ and the result (22), that the joint density of θ and λ will be approximately

$$\{\pi(\lambda)/\rho_0 Q_0\}(Q_1/Q_2)^{\frac{1}{2}} \exp\{-\tfrac{1}{2}n(\theta-\theta_\lambda)^2/\theta_\lambda^2 u^2\},$$

where ρ_0 and the Q_i are evaluated at θ_λ. This is proportional to

$$\theta_\lambda\{\pi(\lambda)/\rho_0 Q_0\}(Q_1/Q_2)^{\frac{1}{2}} \tag{23}$$

times the normal density of θ, given λ. Hence (23) is proportional to the posterior distribution of λ with respect to which the conditional distribution must be integrated to obtain the marginal distribution of θ. We proceed to evaluate (23): ρ_0 is easily found to be

$$(\lambda s_{11}+s_{22})/Q_2-2\lambda/Q_0 = \{\theta^2(s_{22}-\lambda s_{11})+4\theta\lambda s_{12}+\lambda(\lambda s_{11}-s_{22})\}/Q_2 Q_0,$$

and consequently (23) may be written (omitting any constants of proportionality) as $\pi(\lambda)$ times

$$\frac{\theta_\lambda\sqrt{(Q_1 Q_2)}}{(\theta_\lambda^2-4\lambda\theta_\lambda/t-\lambda)(s_{22}-\lambda s_{11})} \quad \propto \frac{\theta_\lambda(\lambda+\theta_\lambda^2)}{(\theta_\lambda^2-4\lambda\theta_\lambda/t-\lambda)(s_{22}-\lambda s_{11})}$$

$$\propto \frac{\lambda+\theta_\lambda^2}{(t+4\lambda/t)(s_{22}-\lambda s_{11})} \quad \propto \frac{\lambda+\theta_\lambda^2}{\{\lambda^2 s_{11}^2+2\lambda(2s_{12}^2-s_{11}s_{22})+s_{22}^2\}}. \tag{24}$$

(The second line in this argument follows on eliminating Q_1 and Q_2 by the same method used in deriving (22).) Denoting this quantity by $\pi^*(\lambda)$, the marginal density of θ is finally obtained as

$$u^{-1}\sqrt{(n/2\pi)}\int \pi^*(\lambda)\,\theta_\lambda^{-1}\exp\{-\tfrac{1}{2}n(\theta-\theta_\lambda)^2/\theta_\lambda^2 u^2\}\,d\lambda, \tag{25}$$

which is best calculated by numerical integration in any particular case.

It may be easier to work in terms of $\log\theta$. Since θ has standard deviation proportional to θ_λ it follows, in the usual way that $\log\theta$ will have approximately constant variance u^2/n. Consequently, (25) can be written

$$u^{-1}\sqrt{(n/2\pi)}\int \pi^*(\lambda)\exp\{-\tfrac{1}{2}n(\log\theta-\log\theta_\lambda)^2/u^2\}\,d\lambda. \tag{26}$$

In other words, to a good approximation $\log\theta-\log\theta_\lambda$ is $N(0, u^2/n)$ independent of θ_λ.

5. MISCELLANEOUS REMARKS

We return to the situation considered at the end of Section 3 where the sample was supposed so large that the variances and covariance of x and y (the θ_{ij}) were effectively known and a simple expression (13) for the posterior density of θ is available. It is clear why the unusual situation of this latter distribution not concentrating around the true value arises: it is because the parameters are not identifiable. That is, the distribution of the observables depends on three parameters only (the θ_{ij}), whereas the problem as formulated depends on four (θ, ϕ_1, ϕ_2 and τ). If a function of the four (for example, $\lambda = \phi_2/\phi_1$) is known, the remaining three can be found; otherwise, not. Previously lack of identifiability has been taken to mean that nothing can be said about the parameters. We argue that this view is false: it often happens, as in the functional relationship situation, that the structure of the problem effectively implies

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

such strong prior knowledge that substantial information about the parameters can be obtained. For example, Moran's inequalities (12) follow from the known non-negative property of the variance. In many applications it will be known that the two errors (in x and y) are of about the same order of magnitude. This might imply that λ lies between k^{-1} and k for suitable k, when quite strong results about the posterior distribution of θ can be made by the methods of Section 4.

It has been shown by Reiersøl (1950) that in the functional relationship problem lack of identifiability occurs only if the distributions are normal. Kiefer and Wolfowitz (1956) similarly show that maximum likelihood is satisfactory provided the distribution of ξ that they use is non-normal. It follows that the results we have obtained are seriously non-robust in the sense that if the ξ-distribution differed from normality by any small amount then the limiting posterior density of θ would be quite different from our form (25) in that it would concentrate about the true value. However, the situation is probably not as bad as this last remark would suggest since the accurate determination of θ would have to exploit the departure from normality and, if this is small, would need a very large sample to be effective. Furthermore, it is possible that an assumption of one type of non-normality, when the true departure from normality is of another form, may lead to inconsistent estimation. What is needed is a procedure that enables a check to be kept on the correctness of the assumptions made. For example, we might make no assumption about the form of the ξ-distribution and estimate it from the observations. Kiefer and Wolfowitz show this is possible, but no method is given: their result is an existence theorem. A Bayesian difficulty here is that such a wide class of distributions cannot be adequately described by a prior distribution. The problem needs much more investigation. Our feeling, based more on intuition than logic, is that the methods we have described are reasonable for distributions of ξ that are not very non-normal, and with moderate sizes of sample. But we may be over-optimistic.

It has been pointed out to us by Professor Daniels that the method of Section 2 may be applied to a problem considered by him and discussed in a paper by Kendall and Babington-Smith (1950). Here $z_i = (x_i, y_i)$ as before, but x_i and y_i are independent normal random variables with means ξ_i and $\xi_i + \theta$ and variances ϕ_1 and ϕ_2 respectively. The means are therefore incidental parameters but their differences are structural. There is no difficulty in estimating θ, but there are snags with ϕ_1 and ϕ_2, as shown in the paper just mentioned. The assignment of a normal distribution to ξ_i with variance τ again yields a bivariate normal distribution for (x_i, y_i) and the estimation is reasonably straightforward. The dispersions are:

$$\text{var}(x_i) = \tau + \phi_1, \quad \text{var}(y_i) = \tau + \phi_2, \quad \text{cov}(x_i, y_i) = \tau.$$

Consequently, the posterior distribution of ϕ_1, for example, concentrates around $s_{11} - s_{12}$ (whose expectation is ϕ_1) and its distribution can be evaluated at least approximately by the methods used in discussing the functional relationship. This problem is identifiable and all the posterior distributions therefore behave in the usual way and concentrate about the true value as the sample size increases. Kendall remarks that this situation is similar to a more complicated one that occurs in factor analysis; and it may be that our method for removing incidental parameters would be useful in that study.

We conclude by making some remarks about assumptions of independence in the prior distribution and about the use of improper prior distributions: that is distributions which do not have a finite integral over the whole range of the parameters.

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

Improper distributions are of considerable use in some situations, particularly where the problem has some group structure: see, for example, Stone (1965). But we have found that in the functional relationship problem they can cause considerable difficulties. This paper arose out of a contribution made by one of us to the discussion on a paper by Sprent (1966). When we came to investigate the remarks made therein with more care we realized that the situation was not as simple as at first assumed. In particular, it had been assumed in the discussion that the incidental parameters were uniformly distributed over the whole real line (an improper distribution) independent of θ. In this form, with the variances known, it led to a result which was unsymmetric in the two variables, x and y. Since the problem has a natural symmetry, this was felt to be undesirable, and instead it was assumed that the radial distance—in the notation of the present paper, $\xi(1+\theta^2)^{\frac{1}{2}}$—was uniform, and independent of θ. This assumption led to a sensible result. In the present paper we assume a proper prior distribution for ξ, namely $N(0, \tau)$, independent of θ, at the cost of including an extra parameter, τ, in the analysis. This leads to sensible answers and the reader may easily verify that the answer is unchanged if, instead, the same assumption had been made about the radial distance. Consequently, the avoidance of the improper prior leads to a desirable invariance of the problem.

Uncritical use of improper distributions can lead to inconsistent estimates. Suppose that we replace ξ by the radial distance when the problem has been scaled so that the two error variances are equal; that is, by $\xi(\phi_1^{-1}+\theta^2\phi_2^{-1})^{\frac{1}{2}}$, and assume this distance to be uniformly distributed. Then the posterior distribution of θ clusters around s_{22}/s_{11} which is not a consistent estimate of θ (compare Lindley, 1947). Again the difficulty is avoided by a proper prior for the incidental parameters. We conjecture that it is always true that when the posterior distribution concentrates it does so about the true value. If it did not we should be in the position of *believing* the parameter to be near to a value which we *knew* not to be the true value: this is hardly an example of consistent beliefs.

It should be noted that where we have used improper priors in this paper (for example, in the integration leading to (15)) these are only in respect of the structural parameters, and even these could be avoided by assuming a sufficiently smooth prior over the effective range of the likelihood function with "tails" sharp enough to ensure the existence of the integral to make the total probability one. More details of the behaviour with other priors will be found in the thesis (El-Sayyad, 1966).

ACKNOWLEDGEMENT

We are extremely grateful to the referee for thorough and helpful comments, in particular in detecting an error in our original maximization of (17).

REFERENCES

ANDERSON, T. W. (1958). *An Introduction to Multivariate Statistical Analysis.* New York: Wiley.

CREASY, MONICA A. (1956). Confidence limits for the gradient in the linear functional relationship. *J. R. Statist. Soc.* B, **18**, 65–69.

EL-SAYYAD, G. M. (1966). Unpublished Ph.D. thesis, University College of Wales, Aberystwyth.

JOHNSTON, J. (1963). *Econometric Methods.* New York: McGraw-Hill.

KENDALL, M. G. and BABINGTON-SMITH, B. (1950). Factor analysis. *J. R. Statist. Soc.* B, **12**, 60–94.

KENDALL, M. G. and STUART, A. (1961). *The Advanced Theory of Statistics*, Volume II. London: Griffin.

KIEFER, J. and WOLFOWITZ, J. (1956). Consistency of the maximum likelihood estimator in the presence of infinitely many incidental parameters. *Ann. Math. Statist.*, **27**, 887–906.

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026

LINDLEY, D. V. (1947). Regression lines and the linear functional relationship. *J. R. Statist. Soc., Suppl.*, **9**, 218–244.

MADANSKY, A. (1959). The fitting of straight lines when both variables are subject to error. *J. Amer. Statist. Ass.*, **54**, 173–205.

MORAN, P. A. P. (1956). A test of significance for an unidentifiable relation. *J. R. Statist. Soc.* B, **18**, 61–64.

NEYMAN, J. and SCOTT, E. L. (1951). On certain methods of estimating the linear structural relation. *Ann. Math. Statist.*, **22**, 352–361.

REIERSØL, O. (1950). Identifiability of a linear relation between variables which are subject to error. *Econometrica*, **18**, 375–389.

SPRENT, P. (1966). A generalized least-squares approach to linear functional relationships. *J. R. Statist. Soc.* B, **28**, 278–297.

STONE, M. (1965). Right Haar measure for convergence in probability to quasi posterior distributions. *Ann. Math. Statist.*, **36**, 440–453.

Downloaded from academic.oup.com/jrsssb/article/30/1/190/7026880 by guest on 08 September 2026